PADHAKU
Ganita Prakash · Class 6 Mathematics

CHAPTER 6

Perimeter and Area

NCERT Exercise Solutions

Figure it Out · 6.1 Perimeter	8 Q
Matha Pachchi! · Figure it Out	9 Q
Estimate and Verify	4 Q
Perimeter of a regular polygon	2 Q
Split and rejoin	2 Q
6.2 Area	1 Q
Figure it Out · 6.2 Area	4 Q
Figure it Out · Tangram	8 Q
Estimating area on squared paper	2 Q
Let's Explore! · Why squares?	4 Q
Let's Explore! · Rectangles of the same area	3 Q
6.3 Area of a Triangle	4 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

Independent study guide for students using the NCERT publication.

Figure it Out · 6.1 Perimeter

1 (a) **Perimeter of a rectangle = 14 cm; breadth = 2 cm; length = ?**

✔ ANSWER

2 × (length + breadth) = 14 cm

length + breadth = 14 ÷ 2 = 7 cm

length = 7 − 2 = **5 cm**

1 (b) **Perimeter of a square = 20 cm; length of a side = ?**

✔ ANSWER

4 × side = 20 cm

side = 20 ÷ 4 = **5 cm**

1 (c) **Perimeter of a rectangle = 12 m; length = 3 m; breadth = ?**

✔ ANSWER

length + breadth = 12 ÷ 2 = 6 m

breadth = 6 − 3 = **3 m**

2 **A rectangle with sides 5 cm and 3 cm is made from a wire. The wire is straightened and bent into a square. What is the length of a side of the square?**

✔ ANSWER

Length of wire = perimeter of rectangle = 2 × (5 + 3) = 16 cm

Side of square = 16 ÷ 4 = **4 cm**

3 **Find the third side of a triangle with perimeter 55 cm and two sides 20 cm and 14 cm.**

✔ ANSWER

Sum of two sides = 20 + 14 = 34 cm

Third side = 55 − 34 = **21 cm**

4 **Find the cost of fencing a rectangular park 150 m long and 120 m wide at ₹40 per metre.**

✔ ANSWER

Perimeter = 2 × (150 + 120) = 2 × 270 = 540 m

Cost = 540 × ₹40 = **₹21,600**

5 **A piece of string is 36 cm long. Find the length of each side if it is used to form: (a) a square, (b) a triangle with all sides equal, (c) a hexagon with all sides equal.**

✔ ANSWER

(a) 36 ÷ 4 = **9 cm**

(b) 36 ÷ 3 = **12 cm**

(c) 36 ÷ 6 = **6 cm**

6 **A farmer's rectangular field is 230 m long and 160 m wide. He fences it with 3 rounds of rope. Find the total length of rope needed.**

✔ ANSWER

Perimeter = 2 × (230 + 160) = 2 × 390 = 780 m

Rope for 3 rounds = 3 × 780 = **2340 m**

Matha Pachchi! · Figure it Out

1 **Akshi runs on the outer track (70 m × 40 m) for 5 rounds. Find the total distance she covered.**

✔ ANSWER

One round = 2 × (70 + 40) = 220 m

5 rounds = 5 × 220 = **1100 m**

2 **Toshi runs on the inner track (60 m × 30 m) for 7 rounds. Find the total distance. Who ran a longer distance?**

✔ ANSWER

One round = 2 × (60 + 30) = 180 m

7 rounds = 7 × 180 = **1260 m**

1260 m > 1100 m, so **Toshi ran the longer distance**.

3 (a) **Mark 'A' where Akshi will be after running 250 m.**

💡 **SAMPLE ANSWER**

One round = 220 m. 250 − 220 = 30 m

Akshi finishes 1 round and goes **30 m past her starting point**. Mark A there (check the picture in your book).

3 (b) **Mark 'B' where Akshi will be after running 500 m.**

💡 **SAMPLE ANSWER**

2 rounds = 440 m. 500 − 440 = 60 m

She is **60 m past her starting point** (after 2 full rounds). Mark B there (check the picture in your book).

3 (c) **Akshi ran 1000 m. How many full rounds has she finished? Mark her position as 'C'.**

💡 **SAMPLE ANSWER**

1000 ÷ 220: 4 rounds = 880 m, 1000 − 880 = 120 m

She has finished **4 full rounds** and is **120 m past her starting point**. Mark C there (check the picture in your book).

3 (d) **Mark 'X' where Toshi will be after running 250 m.**

💡 **SAMPLE ANSWER**

One round = 180 m. 250 − 180 = 70 m

Toshi is **70 m past her starting point**. Mark X there (check the picture in your book).

3 (e) **Mark 'Y' where Toshi will be after running 500 m.**

💡 **SAMPLE ANSWER**

2 rounds = 360 m. 500 − 360 = 140 m

Toshi is **140 m past her starting point**. Mark Y there (check the picture in your book).

3 (f) **Toshi ran 1000 m. How many full rounds has she finished? Mark her position as 'Z'.**

💡 **SAMPLE ANSWER**

5 rounds = 900 m. 1000 − 900 = 100 m

She has finished **5 full rounds** and is **100 m past her starting point**. Mark Z there (check the picture in your book).

Deep Dive **Square tracks: inner side 100 m, outer side 150 m. The common finishing line is at the centre of one side. For a 350 m race, mark the starting points A (inner track) and B (outer track).**

✔ ANSWER

Flag is at the middle of a side, so it is 50 m from a corner on the inner track and 75 m on the outer track.

Inner track: 100 + 100 + 100 + 50 = 350 m. So **A is at a corner**: the runner goes along three full sides and then 50 m to the flag.

Outer track: 125 + 150 + 75 = 350 m. So **B is on the side opposite the finishing line, 125 m before the corner** (25 m from the other corner); the runner then runs one full side (150 m) and 75 m to the flag.

⭐ Estimate and Verify

Activity **Cut random shapes from paper. Estimate the length of each boundary, then measure it with a scale or tape to verify.**

💡 SAMPLE ANSWER

Example: estimate 40 cm, measured 43 cm.

Measure curved edges with a thread and then the scale. With practice, estimates come closer to the measured perimeter.

? **Akshi says the perimeter of this triangle shape is 9 units. Toshi says it will be more than 9 units. What do you think?**

✔ ANSWER

Toshi is right. The slanting (diagonal) side of a grid square is longer than its straight side.

So the perimeter is more than 9 units.

? **Measure a red line and a blue line. Are they the same?**

✔ ANSWER

No. The blue (diagonal) line is longer than the red (straight) line.

? **Write the perimeters of the figures below in terms of straight and diagonal units.**

✔ ANSWER

8s + 2d, **4s + 6d**, **12s + 6d**, **18s + 6d** (in the order of the figures)

⭐ Perimeter of a regular polygon

? **What is a similarity between a square and an equilateral triangle?**

✔ ANSWER

Both have all sides equal and all angles equal, so both are regular polygons.

So the perimeter of each = number of sides × length of a side.

? **Find objects around you with regular shapes and find their perimeters. Generalise for other regular polygons.**

💡 SAMPLE ANSWER

Example: a square tile of side 30 cm → 4 × 30 = 120 cm.

A regular hexagon nut of side 1 cm → 6 × 1 = 6 cm.

General rule: **Perimeter of a regular polygon = number of sides × length of one side.**

⭐ Split and rejoin

? **A 6 cm × 4 cm chit is cut into two equal pieces (6 cm × 2 cm each). Arrangement (a) has perimeter 28 cm. Find the perimeters of arrangements (b), (c) and (d).**

💡 SAMPLE ANSWER

Each piece has perimeter 2 × (6 + 2) = 16 cm; two pieces = 32 cm.

When joined, the touching length is lost from both pieces: perimeter = 32 − 2 × (touching length).

(b) pieces touch along 3 cm: 32 − 6 = **26 cm**

(c) pieces touch along 2 cm: 32 − 4 = **28 cm**

(d) pieces touch along 2 cm: 32 − 4 = **28 cm**

Use the touching length you see in each picture (check the picture in your book).

? **Arrange the two pieces to form a figure with a perimeter of 22 cm.**

✔ ANSWER

We need 32 − 2 × (touching length) = 22, so touching length = 5 cm.

Place one piece on top of the other, along the 6 cm sides, but **shift it by 1 cm** so they touch along 5 cm.

Perimeter = 32 − 10 = **22 cm**

⭐ 6.2 Area

? **Area of a square = ____ ; Area of a rectangle = ____**

✔ ANSWER

Area of a square = **side × side**

Area of a rectangle = **length × width**

Figure it Out · 6.2 Area

1 **The area of a rectangular garden 25 m long is 300 sq m. What is its width?**

✔ ANSWER

Width = area ÷ length = 300 ÷ 25 = **12 m**

2 **Find the cost of tiling a rectangular plot 500 m long and 200 m wide at ₹8 per hundred sq m.**

✔ ANSWER

Area = 500 × 200 = 1,00,000 sq m

Number of hundreds = 1,00,000 ÷ 100 = 1000

Cost = 1000 × ₹8 = **₹8000**

3 **A coconut grove is 100 m long and 50 m wide. Each tree needs 25 sq m. Find the maximum number of trees.**

✔ ANSWER

Area = 100 × 50 = 5000 sq m

Trees = 5000 ÷ 25 = **200**

4 **By splitting the figures into rectangles, find their areas (measures in metres).**

✔ ANSWER

Split each figure into rectangles, find each area (length × width) and add.

(a) **28 sq m**

(b) **9 sq m**

Check the splits with the measures in your book (check the picture in your book).

Figure it Out · Tangram

1 **Explore and find how many pieces have the same area.**

✔ ANSWER

A and B have the same area (big triangles).

C and E have the same area (small triangles).

D, F and G have the same area (each equals two small triangles).

2 **How many times bigger is Shape D than Shape C? What is the relationship between C, D and E?**

✔ ANSWER

D is **2 times** C.

D can be covered exactly by C and E: **area of D = area of C + area of E**.

3 **Which has more area: Shape D or F? Give reasons.**

✔ ANSWER

Both have the same area. Each can be covered exactly by the two small triangles C and E, so each is 2 times C.

4 **Which has more area: Shape F or G? Give reasons.**

✔ ANSWER

Both have the same area. Each is made of two small triangles, i.e., 2 times C.

5 **What is the area of Shape A compared to Shape G? Twice? Four times?**

✔ ANSWER

A = 4 times C and G = 2 times C.

So A is **twice as big** as G.

6 **Find the area of the big square formed with all seven pieces in terms of the area of Shape C.**

✔ ANSWER

A + B = 4 + 4 = 8 C

C + E = 1 + 1 = 2 C

D + F + G = 2 + 2 + 2 = 6 C

Total = 8 + 2 + 6 = **16 times the area of C**

7 **Arrange the 7 pieces into a rectangle. What is its area in terms of Shape C? Give reasons.**

✔ ANSWER

16 times the area of C. The same 7 pieces are used, without gaps or overlaps, so the total area does not change.

8 **Are the perimeters of the square and the rectangle (from the 7 pieces) the same or different? Explain.**

✔ ANSWER

Different. The area is the same, but the shapes are different.

The pieces touch each other in different ways, so the boundary length changes.

The rectangle has a **longer perimeter** than the square (for the same area, a square has the smallest perimeter).

⭐ Estimating area on squared paper

? **Look at figures (a) and (b). Guess which one has a larger area.**

💡 SAMPLE ANSWER

Guess first, then check by counting squares on squared paper using the rules given.

The figure that covers more full squares has the larger area (check the picture in your book).

? **Find the area of the following figures (on squared paper).**

✔ ANSWER

Count full squares as 1, more than half as 1, exactly half as ½, less than half as 0.

Areas: **4 sq units, 9 sq units, 10 sq units, 11 sq units** (in the order of the figures)

⭐ Let's Explore! · Why squares?

? **Draw a circle of diameter 3 units on a graph sheet. Count the squares and estimate its area.**

💡 SAMPLE ANSWER

Count full squares and parts more than half.

The area comes to **about 7 sq units**.

? **Try triangles and rectangles to fill a space. List points that make a square the best shape to measure area.**

💡 SAMPLE ANSWER

- Squares fit together with **no gaps and no overlaps**.
- All sides are equal, so a square is the same in every direction.
- Squares line up in rows and columns, so counting is easy (length × width).
- A square of side 1 unit gives a simple unit: 1 sq unit.
- Circles leave gaps, and triangles must be turned around, so counting is harder.

1 **Find the area (in sq m) of the floor outside the corridor.**

💡 **SAMPLE ANSWER**

Measure the length and width with a measuring tape.

Example: 20 m × 3 m → area = 20 × 3 = **60 sq m**

2 **Find the area (in sq m) of your school playground.**

💡 **SAMPLE ANSWER**

Measure its length and width (in steps or with a tape).

Example: 50 m × 40 m → area = 50 × 40 = **2000 sq m**

⭐ Let's Explore! · Rectangles of the same area

a, b **Make rectangles with whole-number sides and area 24 sq units. Which has the greatest perimeter? The least?**

✔ ANSWER

Length × width	Perimeter
24 × 1	50 units
12 × 2	28 units
8 × 3	22 units
6 × 4	20 units

(a) Greatest perimeter: **24 × 1** (50 units)

(b) Least perimeter: **6 × 4** (20 units)

c **If the area is 32 sq cm, what will your answers be?**

✔ ANSWER

Length × width	Perimeter
32 × 1	66 cm
16 × 2	36 cm
8 × 4	24 cm

Greatest: **32 cm × 1 cm** (66 cm); least: **8 cm × 4 cm** (24 cm)

? **Given any area, can you predict the rectangle with the greatest and the least perimeter?**

✔ ANSWER

Yes.

- The longest, thinnest rectangle (width 1) has the **greatest** perimeter.
- The rectangle closest to a square has the **least** perimeter.

Example: area 36 → 36 × 1 has perimeter 74; 6 × 6 has perimeter 24.

⭐ 6.3 Area of a Triangle

? **Cut a rectangle along a diagonal. Do the two triangles overlap exactly? Do they have the same area? What can you infer?**

✔ ANSWER

Yes, the two triangles overlap exactly, so they have the **same area**.

Inference: a diagonal divides a rectangle (or square) into two equal triangles, so **each triangle has half the area of the rectangle**.

? **Is the area of the blue rectangle more or less than the area of the yellow triangle, or the same? Why? Write the relationship.**

💡 SAMPLE ANSWER

The triangle has the same base and height as the rectangle.

The triangle covers exactly half of the rectangle, so **area of triangle = ½ × area of rectangle** (the rectangle has twice the area) (check the picture in your book).

1, 2 **Using grid paper, find the area of blue triangle BAD and red triangle ABE. Area of rectangle ABCD = ____.**

💡 **SAMPLE ANSWER**

Area of rectangle ABCD = AB × AD (count the squares).

Triangle BAD is half of rectangle ABCD.

Triangle ABE = half of AFED + half of BFEC = half of ABCD.

So **both triangles have the same area = ½ × area of ABCD** (check the picture in your book).

Conclusion **Write the conclusion.**

✔ **ANSWER**

A triangle drawn on the side of a rectangle, with its third vertex on the opposite side, has half the area of the rectangle.

Two triangles with the same base and the same height have the same area, even if they look different.

Figure it Out · 6.3 Area of a Triangle

1 **Find the areas of the figures by dividing them into rectangles and triangles.**

✔ ANSWER

Split each figure into rectangles and triangles; a triangle is half of its rectangle; add all parts.

(a) **24 sq units**

(b) **30 sq units**

(c) **48 sq units**

(d) **16 sq units**

(e) **12 sq units**

Making it 'More' or 'Less'

1 **Using 9 unit squares, what is the smallest perimeter possible?**

✔ ANSWER

Arrange them as a 3 × 3 square.

Perimeter = 4 × 3 = **12 units**

2 **What is the largest perimeter possible?**

✔ ANSWER

Place every square so that it shares only one side with the figure, e.g., all 9 in a row (9 × 1).

Perimeter = 2 × (9 + 1) = **20 units**

3 **Make a figure with a perimeter of 18 units.**

✔ ANSWER

Each square has 4 sides: 9 × 4 = 36. Every shared side removes 2 units.

We need 36 − 18 = 18, i.e., 9 shared sides.

Example: a row of 7 squares with 2 squares placed on top of the first two.

Shared sides = 6 + 2 + 1 = 9 → perimeter = **18 units**

4 **Can you make other shapes for each of these perimeters, or only one? Why?**

✔ ANSWER

- 12 units: **only one shape**, the 3 × 3 square.
- 20 units: **many shapes** (straight line, L-shape, zig-zag...), any shape with no 2 × 2 block.
- 18 units: **many shapes**, any shape with exactly 9 shared sides.

Reason: the perimeter depends only on how many sides are shared, and that can happen in many ways.

? **A figure has perimeter 24 units. What will the new perimeter be if a new square is attached as shown?**

💡 SAMPLE ANSWER

A new square adds 4 sides but hides the sides it touches (on both shapes).

Touching 1 side: perimeter **increases by 2** → 26 units.

Touching 2 sides: **stays the same** → 24 units.

Touching 3 sides: **decreases by 2** → 22 units.

Count how many sides the new square touches in the picture (check the picture in your book).

a, b, c **Place the square so that the perimeter (a) increases; (b) decreases; (c) stays the same.**

✔ ANSWER

(a) Place it touching **only one side** → perimeter increases by 2.

(b) Place it in a gap where it touches **three sides** → perimeter decreases by 2.

(c) Place it in a corner where it touches **two sides** → perimeter stays the same.

⭐ House plans

? **Charan's house plan (rectangular plot, height 30 ft). Master Bedroom 15 ft × 15 ft = 225 sq ft; Kitchen 15 ft × 12 ft = 180 sq ft; Toilet 5 ft × 10 ft. (a) Find the missing measurements. (b) Find the area of his house.**

✔ ANSWER

(a) Missing measurements:

Room	Size	Area
Small Bedroom	15 ft × 12 ft	180 sq ft
Utility	15 ft × 3 ft	45 sq ft
Hall	20 ft × 12 ft	240 sq ft
Parking	15 ft × 3 ft	45 sq ft
Garden	20 ft × 3 ft	60 sq ft

(b) The plot is 35 ft × 30 ft.

Area = 35 × 30 = **1050 sq ft**

? **Sharan's house plan (width 42 ft). Kitchen 18 ft × 10 ft = 180 sq ft; Master Bedroom 12 ft × 15 ft = 180 sq ft; Small Bedroom 12 ft × 10 ft; Hall 23 ft × __; a room of area 70 sq ft. (a) Find the missing measurements. (b) Find the area of his house.**

✔ ANSWER

(a)

Room	Size	Area
Utility	7 ft × 10 ft	70 sq ft
Toilet	5 ft × 10 ft	50 sq ft
Kitchen	18 ft × 10 ft	180 sq ft
Master Bedroom	12 ft × 15 ft	180 sq ft
Hall	23 ft × 15 ft	345 sq ft
Entrance	7 ft × 15 ft	105 sq ft
Small Bedroom	12 ft × 10 ft	120 sq ft

(b) The plot is 42 ft × 25 ft (10 + 15 = 25).

Area = 42 × 25 = **1050 sq ft**

Check: 70 + 50 + 180 + 180 + 345 + 105 + 120 = 1050 sq ft

? **Compare the areas and perimeters of Sharan's house and Charan's house.**

✔ ANSWER

Areas: Charan 35 × 30 = 1050 sq ft; Sharan 42 × 25 = 1050 sq ft → **areas are equal**.

Perimeters: Charan 2 × (35 + 30) = 130 ft; Sharan 2 × (42 + 25) = 134 ft.

Sharan's house has the greater perimeter, though both have the same area.

⭐ Area Maze Puzzles

a **Find the missing area. (Regions 13 sq cm and 26 sq cm on top; 15 sq cm and ? below.)**

✔ ANSWER

Same height: 26 is 2 times 13, so the right width is 2 times the left width.

Below, the same widths: ? = 2 × 15 = **30 sq cm**

b **Find the missing area.**

✔ ANSWER

Use the given sides and areas: width = area ÷ length, step by step.

Missing area = 3 cm × 3 cm = **9 sq cm**

c **Find the missing area.**

💡 **SAMPLE ANSWER**

Find unknown sides using side = area ÷ other side, then multiply.

Missing area = **16 sq cm** (check the picture in your book).

d **Find the missing length.**

💡 **SAMPLE ANSWER**

Find unknown sides using side = area ÷ other side (e.g., 42 ÷ 6 = 7 cm).

Missing length = **5 cm** (check the picture in your book).

🧮 Figure it Out · Perimeter and Area

1 **Give the dimensions of a rectangle whose area is the sum of the areas of rectangles 5 m × 10 m and 2 m × 7 m.**

✔ **ANSWER**

Area = 5 × 10 + 2 × 7 = 50 + 14 = 64 sq m

Possible rectangles: **8 m × 8 m**, **16 m × 4 m**, **32 m × 2 m** or **64 m × 1 m**

2 **A rectangular garden 50 m long has area 1000 sq m. Find its width.**

✔ ANSWER

Width = 1000 ÷ 50 = **20 m**

3 **A floor is 5 m long and 4 m wide. A square carpet of side 3 m is laid on it. Find the area not carpeted.**

✔ ANSWER

Floor = 5 × 4 = 20 sq m

Carpet = 3 × 3 = 9 sq m

Not carpeted = 20 − 9 = **11 sq m**

4 **Four flower beds 2 m × 1 m are dug at the corners of a garden 15 m × 12 m. How much area is left for a lawn?**

✔ ANSWER

Garden = 15 × 12 = 180 sq m

Flower beds = 4 × (2 × 1) = 8 sq m

Lawn = 180 − 8 = **172 sq m**

5 **Shape A has area 18 sq units, Shape B has area 20 sq units, and A has a longer perimeter than B. Draw two such shapes.**

💡 SAMPLE ANSWER

Shape A: rectangle **18 × 1** → perimeter 2 × (18 + 1) = 38 units

Shape B: rectangle **5 × 4** → perimeter 2 × (5 + 4) = 18 units

38 > 18, so A has the longer perimeter. Draw them on squared paper.

6 **On a page of your book, draw a rectangular border 1 cm from the top and bottom and 1.5 cm from the left and right. What is its perimeter?**

💡 **SAMPLE ANSWER**

Border length = page length − 1 − 1; border width = page width − 1.5 − 1.5

Example: page 28 cm × 21 cm → border 26 cm × 18 cm

Perimeter = 2 × (26 + 18) = **88 cm**

(In general, it is 10 cm less than the perimeter of the page.)

7 **Draw a 12 units × 8 units rectangle. Draw another rectangle inside it, not touching it, with exactly half the area.**

✔ **ANSWER**

Outer area = 12 × 8 = 96 sq units; half = 48 sq units

Inner rectangle: **8 units × 6 units** (8 × 6 = 48)

Place it in the middle: 2 units gap on the left and right, 1 unit gap at top and bottom.

8 **A square paper is folded in half and cut along the fold into two rectangles. Which statement is always true?**

💡 **SAMPLE ANSWER**

(a) Each rectangle has more area than the square.

(b) Perimeter of the square > sum of the perimeters of both rectangles.

(c) Sum of the perimeters of both rectangles = 1½ times the perimeter of the square.

(d) Area of the square = 3 times the areas of both rectangles added.

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**